

Data Card: Complaint Sense – Consumer Complaint Classification Dataset

PROBLEM

Consumer complaints contain important information about the problems people experience with products and services, but manually reviewing and categorizing large numbers of complaints can be time-consuming. Automated text classification can help organize complaints by identifying the issue described in the complaint narrative. This dataset supports research on classifying real-world consumer financial complaints based on consumer language. The dataset is sourced from the Consumer Financial Protection Bureau (CFPB) Consumer Complaint Database, which provides publicly available complaint data concerning financial products and services. Complaint narratives are written by consumers and may contain informal language, different writing styles, and varying levels of detail. The dataset can support research on automated complaint routing, complaint analysis, customer service support, and natural language processing. However, it primarily reflects the U.S. financial services environment and English-language complaints, so it does not fully capture multilingual or non-U.S. consumer experiences.

Dataset Link

CFPB Consumer Complaint Database — <https://www.consumerfinance.gov/data-research/consumer-complaints/>

Data Card Authors

- Nkwor Jane Chinelo, Team: Stanley (Contributor)
- Oke Oluwajomiloju, Team: Stanley (Contributor)

Input vs. Output	
• What are users inputting? • What are the outputs or labels your model will predict or generate?	Input: Text-based consumer complaint narratives. Output: A complaint category based on the selected CFPB Issue labels.

Classification Problem	
• What is the classification task, if any? • What categories or classes are you working with, and why?	This is a multi-class text classification task. The model will classify consumer complaint narratives into 11 selected CFPB Issue categories. These categories were selected after exploratory analysis to prioritize clear complaint types with sufficient representation while excluding the vague categories “Other features, terms, or problems” and “Dealing with your lender or servicer.”

Labeling	
• Where do your labels come from—domain experts, community members, crowd workers, AI-generated sources? • How will you check for consistency and accuracy?	Labels come from the existing Issue classifications provided by the CFPB Consumer Complaint Database. We will inspect the label distribution, remove exact duplicates to reduce leakage, review representative narratives for each selected category, and exclude unsuitable or ambiguous categories.

Data Creation	
• How will you collect your data? • Will it come from direct sources (like interviews or recordings), or will you synthesize data in low-resource settings?	The primary data source is the official CFPB Consumer Complaint Database. For this working dataset, complaints from 2025-08-16 through 2026-08-16 with published narratives were exported and filtered. The source is direct, real-world complaint data; no synthetic complaint generation is used for the primary dataset. Language: English.

Web Collection Ethics	
• Do you plan to source data from the web? • If yes, which sites or platforms? • What tools will you use—AI tools, human workers, or both? • Have you addressed issues of consent, copyright, and privacy?	Yes. Source: Consumer Financial Protection Bureau (CFPB) Consumer Complaint Database. We use the official dataset plus human selection, cleaning, preprocessing, and quality checks; AI tools may assist with analysis and model development, but will not be used to present generated complaints as original CFPB data. The CFPB states that the applicable U.S. Government work is in the public domain in the United States and waives copyright and related rights worldwide through CC0 1.0 Universal, subject to stated exceptions. The CFPB also states that complaint narratives are published when consumers opt in and that reasonable steps are taken to remove identifying personal information. We will not intentionally reintroduce personally identifiable information.

Bias and Representation	
• Who might be left out of your dataset? • How will you ensure that the dataset reflects the diversity of the community and includes underrepresented voices?	The dataset may underrepresent consumers who do not submit CFPB complaints, consumers outside the CFPB's financial-services scope, non-U.S. consumers, non-English speakers, and people who use local languages or code-switch. These limitations will be documented, class distribution monitored, and future localized data considered for Nigerian English, Nigerian Pidgin, and indigenous Nigerian languages where suitable verified sources are available.

Values and Community Alignment	
• How does your dataset design reflect the values identified in 01 Build Your Own Small Language Model? • How will your community benefit from this dataset, and how will you ensure accountability to them?	Accessibility: the public source supports reproducible NLP research. Fairness: representation gaps, class imbalance, and dataset limitations will be documented and monitored. Privacy: CFPB applies consent and privacy protections to published narratives; preprocessing will avoid retaining unnecessary personal metadata. Transparency: source, preprocessing, target selection, and limitations will be documented. Community benefit: the work can support research toward more localized complaint-classification systems while making limitations clear.

DATASET OVERVIEW

Field	Details
Title	CFPB Consumer Complaint Classification Working Dataset

Task	Multi-class text classification
Domain	Consumer financial complaints
Language	English
Source	Consumer Financial Protection Bureau
License/Access	CC0 1.0 Universal (public-domain dedication), subject to stated exceptions
Total size	1,293 exported complaints; 1,166 unique narratives after exact-duplicate removal; 323 complaints in the current balanced working subset
Metric	Macro F1
Baseline	Pending the first baseline model experiment

Schema

Fields
Complaint ID
Consumer complaint narrative
Product
Sub-product
Issue
Sub-issue
Company
Date received
State/ZIP code
Company response to the consumer
Submitted via

CATEGORIES

CATEGORY	COUNT	EXAMPLE COMPLAINTS
Incorrect information on your report	30	Inaccurate or incorrect information was reported about the consumer.
Improper use of your report	30	Complaint about improper use of a consumer report.
Managing an account	30	Problems managing or using a financial account.
Problem with a company's investigation into an existing problem	30	Complaint that an investigation did not resolve an existing problem.
Problem with a purchase shown on your statement	30	Dispute about a purchase appearing on a statement.
Attempts to collect debt not owed	30	Debt collection for a debt the consumer says is not owed.
Trouble during the payment process	30	Problems with making, processing, or handling payments.
False statements or representation	30	Claims that a company or collector made false statements or representations.
Struggling to pay the mortgage	30	Difficulty making mortgage payments or obtaining payment assistance.
Written notification about debt	30	Problems with written information or notices concerning a debt.
Took or threatened to take negative or legal action	23	Threats or actions involving lawsuits, credit damage, or other legal consequences.

Working-set note: the source extract contains 1,293 complaints. After exact-duplicate removal, the current 11-category working pool contains 806 unique narratives. The balanced modeling subset contains 323 complaints: 30 examples per class where available, and 23 examples for the smallest selected class.

Evaluation

- Primary metric: Macro F1, giving each selected class equal weight.
- Additional metrics: Accuracy, weighted F1, precision, recall, and per-class F1.
- Evaluate on a held-out test set.
- Check duplicate/near-duplicate narratives to reduce leakage; exact duplicate narratives were removed from the working subset before balancing.
- Report class imbalance and document the balancing strategy used for the modeling subset.

Limitations

- English-language complaints; multilingual representation is limited.
- U.S.-focused financial marketplace; not Nigerian/global.
- CFPB states the database is not a statistical sample and is not necessarily representative of all consumers.
- Narratives are available only when consumers opt in to publication.

- Some records may have missing narratives/fields.
- Categories may change over time; document dataset version/extraction date.
- Real-world text can contain spelling errors, informal language, ambiguity, and varying detail.
- Initial system focuses on single-label classification using the selected Issue categories.
- Nigerian Pidgin and indigenous Nigerian languages are not included in the current training data.

Dataset Status

Field	Current status
Primary dataset	CFPB Consumer Complaint Database (working extract)
Training language	English
Multilingual training	Not included in the current working subset
Local-language snippets	Contextual evidence only unless a verified dataset/source is added
Final target category	Selected CFPB Issue label
Final dataset size	323 complaints in balanced working subset
Baseline score	Pending the first baseline experiment
Dataset download date/version	2026-08-16; filtered period 2025-08-16 through 2026-08-16